Inverted Index and Search with Hadoop and Spark

Cloud Computing Project Report

Reza Almassi – Ronald Omoding – Rojan Shrestha

Academic Year 2024/2025

Source code: https://github.com/Rezacs/Cloud

1 Project Overview

The goal of the project was to build a basic search-engine backend based on an *inverted index*. For each word detected in a collection of text files, the output stores the list of filenames where the word appears and its frequency inside each file, for example:

```
cloud  doc1.txt:3 doc2.txt:1
```

We implemented and evaluated five solutions:

1. **Hadoop Base** in Java, using a standard combiner.
2. **Hadoop InMapper** in Java, using in-mapper combining.
3. **PySpark** (Spark Optimized), in Python.
4. **Java Spark**, an equivalent Spark implementation in Java.
5. **Sequential Python**, used as a single-machine baseline.

The distributed Hadoop and Spark methods read input from **HDFS**. The sequential Python solution reads from the **local filesystem**. This is intentional: the baseline uses normal Python APIs such as `os.walk()` and `open()` and is interpreted as a local single-machine reference rather than a cluster job.

1.1 Cluster and Software

The project was deployed on a three-node fully distributed Hadoop cluster using **Hadoop 3.1.3**, **Java 8**, and **Apache Spark**. Each VM had 2 CPU cores and 6.8 GB of RAM. The main node also participated as a worker, running both master-side and worker-side YARN daemons.

Table 1: Cluster node roles.

Node	Role	IP address
`namenode`	NameNode, SecondaryNameNode, ResourceManager, DataNode, NodeManager	`10.1.1.166`
`datanode2`	DataNode, NodeManager	`10.1.1.213`
`datanode3`	DataNode, NodeManager	`10.1.1.163`

HDFS was configured with replication factor 2. YARN was configured with three active NodeManagers, so all three machines were eligible to execute distributed tasks.

2 Dataset Preparation

2.1 Source Collections

The original source files were stored under `/home/hadoop/Cloud/source_downloads/` and came from three collections:

- `gutenberg_big` — public-domain books from Project Gutenberg.
- `archive_big` — text files from the Internet Archive.
- `kaggle_news` — news articles from a Kaggle dataset. The `kaggle_news` source folder was created from the Kaggle News Articles Corpus dataset: https://www.kaggle.com/datasets/sbhatti/news-articles-corpus.

2.2 General Preparation Method

Most datasets were built by selecting files from the source folders using a *round-robin* strategy, taking files alternately from the available collections until a target size was reached (measured in binary megabytes: $1\,\text{MB} = 1024 \times 1024$ bytes). Files were *moved* rather than copied in order to avoid duplication and track remaining source stock. Some later datasets were built by copying or combining existing prepared datasets to create specific combined workloads.

2.3 Individual Dataset Construction

- **ds-00** (100 files, 0.43 MB): a very small news dataset, pre-prepared.
- **ds-01** (1,000 files, 4.14 MB): a small news dataset, pre-prepared.
- **ds-02** (2,582 files, 261 MB): files from all three collections, round-robin.
- **ds-03** (642 files, 297.89 MB): files from `gutenberg_big` and `archive_big` only.
- **ds-04** (844 files, 339 MB): files from all three collections, round-robin.
- **ds-05** (807 files, 500 MB): files from all three collections, round-robin.

- **ds-06** (266 files, 752.03 MB): all remaining `gutenberg_big` files; Gutenberg only.
- **ds-07** (2,493 files, 800 MB): files from all three collections, round-robin.
- **ds-08** (6,680 files, 1126.40 MB): files from all three collections, round-robin.
- **ds-09** (18,680 files, 1173.86 MB): a copy of ds-08 plus 12,000 files from `kaggle_news`.
- **ds-10** (1,917 files, 1591.03 MB): a combination of ds-04, ds-05, and ds-06.
- **ds-11** (116 files, 96 MB): files from the Internet Archive collection.
- **ds-12** (2,495 files, 1126.40 MB): files from Project Gutenberg.

After construction, `gutenberg_big` and `archive_big` were fully consumed. Approximately 850,638 files (≈3.2 GB) remained in `kaggle_news`.

2.4 Combined Datasets

In addition to the 13 individually prepared datasets, 12 *combined* datasets were created by passing multiple comma-separated HDFS paths directly to Hadoop and Spark jobs, without physically copying or merging any files on HDFS. Both `FileInputFormat` (Hadoop) and `wholeTextFiles()` (Spark) accept multiple input directories natively. The sequential Python baseline, which has no multi-path support, instead copies each path into its own subfolder of a temporary local directory and walks the parent recursively.

Table 2: All 25 datasets used in the final experiments, ordered by approximate size.

Dataset	Composition	Files	Approx. Size
ds-00-000mb-100files-tiny-news	base	100	0.43 MB
ds-01-004mb-1000files-news-small	base	1,000	4.14 MB
combo-06-ds00-01	ds-00 + ds-01	1,100	4.57 MB
ds-11-096mb-116files-archive	base	116	96 MB
ds-03-298mb-642files	base	642	297.89 MB
ds-02-261mb-2582files	base	2,582	261 MB
ds-04-339mb-844files	base	844	339 MB
ds-05-500mb-807files	base	807	500 MB
combo-01-ds02-03	ds-02 + ds-03	3,224	559 MB
ds-06-752mb-266files-gutenberg-remaining	base	266	752.03 MB
ds-07-800mb-2493files	base	2,493	800 MB
combo-02-ds06-11	ds-06 + ds-11	382	848 MB
ds-08-1p1gb-6680files	base	6,680	1126.40 MB
ds-12-1p1gb-2495files-gutenberg	base	2,495	1126.40 MB
combo-07-ds03-04-05	ds-03+ds-04+ds-05	2,293	1137 MB
ds-09-1p15gb-18680files	base	18,680	1173.86 MB
combo-04-ds01-09	ds-01 + ds-09	19,680	1178 MB
combo-08-ds02-09	ds-02 + ds-09	21,262	1435 MB
ds-10-1p55gb-1917files-combined	base	1,917	1591.03 MB
combo-09-ds06-07-11	ds-06+ds-07+ds-11	2,875	1648 MB
combo-03-ds07-12	ds-07 + ds-12	4,988	1926 MB
combo-05-ds09-10	ds-09 + ds-10	20,597	2765 MB
combo-11-ds06-07-10	ds-06+ds-07+ds-10	4,676	3142 MB
combo-10-ds08-09-12	ds-08+ds-09+ds-12	27,855	3426 MB
combo-12-ds08-10-12	ds-08+ds-10+ds-12	11,092	3843 MB

3 Implementation

3.1 Common Processing Logic

All implementations share the same tokenization rule (`[a-z0-9]+`), lowercase conversion, and stop-word removal. The output format is:

```
word   filename1:count1 filename2:count2 ...
```

3.2 Hadoop Base with Combiner

The Mapper emits one key-value pair per non-stopword token, keyed as `word@filename`. A Combiner sums equal keys before the shuffle, and the Reducer converts all `word@filename` pairs into one posting list per word. A custom `WholeFileInputFormat` reads each file as a single record using the filename as the map key, which is necessary to preserve original filenames in the output.

Listing 1: Hadoop Base: mapper output and combiner.

```
// Mapper
while (matcher.find()) {
    if (!stopWords.contains(matcher.group())) {
        outputKey.set(matcher.group() + "@" + filename);
        context.write(outputKey, one);
    }
}
// Combiner
int sum = 0;
for (IntWritable v : values) sum += v.get();
context.write(key, new IntWritable(sum));
```

3.3 Hadoop InMapper Combining

The InMapper variant accumulates word counts in a per-file `HashMap` inside the mapper and emits only one record per unique `word@filename` pair. This eliminates repeated intermediate records for the same word within a file, directly reducing shuffle volume and reducer pressure.

Listing 2: Hadoop InMapper: local aggregation.

```
localCounts.clear();
while (matcher.find()) {
    String word = matcher.group();
    if (!stopWords.contains(word)) {
        String k = word + "@" + filename;
        localCounts.put(k, localCounts.getOrDefault(k, 0) + 1);
    }
}
for (Map.Entry<String,Integer> e : localCounts.entrySet())
    context.write(new Text(e.getKey()), new IntWritable(e.getValue()));
```

3.4 PySpark (Spark Optimized)

The PySpark solution uses `wholeTextFiles()` so each file is processed as a single document. Words are counted locally per file, then `combineByKey` aggregates posting lists across partitions. The global `sortByKey()` was removed because it introduced an additional wide shuffle with no benefit for the inverted index itself. Postings within each word's list are still sorted locally.

Listing 3: PySpark core pipeline.

```
files = sc.wholeTextFiles(input_path, minPartitions=num_partitions * 2)
inverted_index = (
    files
    .flatMap(file_to_postings) # local per-file word counting
    .combineByKey(create_combiner,
                  merge_value,
                  merge_combiners,
                  numPartitions=num_partitions)
    .mapValues(lambda postings: " ".join(sorted(postings)))
)
inverted_index.map(lambda x: f"{x[0]}\t{x[1]}").saveAsTextFile(output_path)
```

Additional optimizations enabled: Kryo serialization, shuffle compression, RDD compression, and Python worker reuse (`spark.python.worker.reuse=true`).

3.5 Java Spark

A second Spark implementation was written in Java, using identical algorithmic structure to PySpark (per-file local counting, `combineByKey`, no global sort) so that the language and runtime overhead can be isolated from algorithmic behaviour. Stop-words are broadcast to all executors via `JavaSparkContext.broadcast()`.

Listing 4: Java Spark: flatMapToPair and combineByKey.

```
JavaPairRDD<String,String> postings = files.flatMapToPair(file -> {
    String filename = file._1().substring(file._1().lastIndexOf("/") + 1);
    Map<String,Integer> counts = new HashMap<>();
    Matcher m = TOKEN_RE.matcher(file._2());
    while (m.find()) {
        String w = m.group().toLowerCase();
        if (!stopwordsBC.value().contains(w))
            counts.put(w, counts.getOrDefault(w, 0) + 1);
    }
    List<Tuple2<String,String>> out = new ArrayList<>();
    for (Map.Entry<String,Integer> e : counts.entrySet())
        out.add(new Tuple2<>(e.getKey(), filename + ":" + e.getValue()));
    return out.iterator();
});

JavaPairRDD<String,List<String>> index = postings.combineByKey(
    v -> { List<String> l = new ArrayList<>(); l.add(v); return l; },
    (l,v) -> { l.add(v); return l; },
    (a,b) -> { a.addAll(b); return a; },
    partitions
);
index.map(t -> { Collections.sort(t._2());
                 return t._1() + "\t" + String.join(" ", t._2()); })
     .saveAsTextFile(output);
```

3.6 Spark Cluster Configuration

Both Spark engines were submitted with the same fixed YARN configuration for every dataset, removing executor memory as a confounding variable between datasets.

Table 3: Spark submit configuration, applied uniformly to all 25 datasets and both Spark engines.

Parameter	Value
`-master` / `-deploy-mode`	`yarn` / `client`
`-num-executors`	3
`-executor-memory`	3584 m
`-executor-cores`	2
`spark.executor.memoryOverhead`	512 m
`-driver-memory`	2 g (on `namenode`, outside YARN cap)
`spark.yarn.am.memory`	512 m
`spark.dynamicAllocation.enabled`	false
`spark.hadoop.mapreduce.input.fileinputformat.list-status.num-threads`	8
`spark.network.timeout`	600 s
`spark.executor.heartbeatInterval`	60 s

Each executor requests $3584 + 512 = 4096$ MB; three executors plus the Application Master ($512 + \approx 384$ MB overhead) total approximately 13.2 GB out of the cluster's 16.2 GB YARN capacity. Setting `executor-cores=2` allows each executor JVM to use both physical cores of its node, replacing an earlier single-core configuration.

3.7 Sequential Python and Search

The sequential baseline builds the inverted index in a Python `defaultdict` and writes output in sorted word order from local files only.

Listing 5: Sequential Python core loop.

```
inverted_index = defaultdict(dict)
for filepath in collect_files(input_path):
    filename = os.path.basename(filepath)
    with open(filepath, "r", encoding="utf-8", errors="replace") as f:
        text = f.read()
    counts = Counter(w for w in tokenize(text) if w not in stopwords)
    for word, count in counts.items():
        inverted_index[word][filename] = count
```

A non-parallel search script loads the generated index and answers single-term or multi-term queries, returning the intersection of filename sets for multi-term queries.

4 Experimental Methodology

4.1 Cluster Resource Configuration

Table 4: Per-node and cluster-wide YARN resource configuration.

Parameter	Value
RAM per node (physical)	6.8 GB
CPU cores per node (physical)	2
`yarn.nodemanager.resource.memory-mb`	5,400 MB
`yarn.nodemanager.resource.cpu-vcores`	3
`yarn.scheduler.maximum-allocation-mb`	5,400 MB
Cluster-wide YARN memory (3×5400)	16,200 MB
Cluster-wide YARN vcores (3×3)	9
HDFS replication factor	2

The configured 3 vcores per node slightly exceeds the 2 physical cores available; this pre-existing oversubscription was left unchanged.

4.2 Parallelism Parameters

For **Hadoop** (both Base and InMapper), the reducer count was swept as $r \in \{1, 2, 4, 8, 16, 24\}$. Hadoop's disk-backed shuffle architecture does not suffer from single-reducer memory collapse, so $r = 1$ remained a valid and informative measurement across all 25 datasets.

For **Spark** (both PySpark and Java Spark), the partition count was swept as $p \in \{4, 8, 16, 24, 32, 40\}$. The values $p = 1$ and $p = 2$ used in earlier runs were removed because they consistently caused either immediate job failure (`exit status 1`, no output) or runtimes of 30–50 minutes on datasets larger than $\approx$1 GB: with a single partition, the entire dataset must be held in one executor's heap during garbage collection, which exhausts the available memory. The values $p = 32$ and $p = 40$ were added to determine whether performance continued to improve beyond the $p = 24$ plateau observed in earlier runs.

4.3 Measurement Procedure

Each job was launched under `/usr/bin/time -v` to record wall-clock elapsed time and the peak resident set size (RSS) of the client/driver process. A background monitor sampled `yarn node -status` and `free -m` on all three nodes every 5 seconds throughout each job, recording per-node YARN-allocated memory and per-node OS memory usage. The per-node

samples were summed across all three nodes at each timestamp; the maximum across all timestamps was recorded as the job's peak cluster-wide system memory (`system_used_max_gb`) and peak YARN allocation (`max_yarn_allocated_gb`). HDFS output was deleted and `/tmp` was cleared on all three nodes between every individual job run, ensuring no residual data could affect subsequent measurements. Correctness was validated by counting output lines for every job.

4.4 Anomalies Encountered and Resolved

ds-09 nested HDFS directory. In an earlier run, all Hadoop jobs on `ds-09` produced zero output lines despite exit status 0, and all Spark jobs failed immediately after 30–50 seconds. Inspection revealed that all 18,680 files were nested one directory level deeper than expected due to how the dataset was uploaded. Hadoop's `FileInputFormat` does not recurse into subdirectories by default, so it found zero input splits and produced a technically successful but empty result; Spark raised an explicit "0 files matched" error at job-planning time. The fix was a one-time `hdfs dfs -mv` to flatten the directory. After the fix, all five methods produced the expected output.

combo-11 sequential Python out-of-memory. The sequential Python run on `combo-11` (3.1 GB) was terminated by the OS out-of-memory killer (exit status 137) after 423 seconds. Holding the full in-memory dictionary for 3.1 GB of input on a single machine with no spill-to-disk mechanism exceeded available RAM on the `namenode`. All four distributed methods completed `combo-11` successfully.

5 Results

Out of 625 individual job executions, 624 completed successfully. The one exception (combo-11 sequential Python, OOM) is excluded from the tables below.

5.1 Best Elapsed Time per Dataset

Table 5: Best elapsed time (seconds) per dataset and method. r = reducer count for Hadoop; p = partition count for Spark. Sequential Python is a single local run with no parameter. `--` = OOM failure.

Dataset	Size (MB)	H.Base	H.InMap	PySpark	J.Spark	Seq.Py
ds-00 (0.43 MB, 100 files)	0.43	29	25	41	39	0
ds-01 (4.14 MB, 1000 files)	4.14	35	32	42	42	1
combo-06 (4.57 MB, 1100 files)	4.57	33	31	42	40	2
ds-11 (96 MB, 116 files)	96	73	45	48	46	17
ds-03 (297.89 MB, 642 files)	297.89	119	70	61	61	52
ds-02 (261 MB, 2582 files)	261	127	78	62	62	48
ds-04 (339 MB, 844 files)	339	116	67	59	63	58
ds-05 (500 MB, 807 files)	500	98	66	70	72	84
combo-01 (559 MB, 3224 files)	559	128	82	81	78	100
ds-06 (752 MB, 266 files)	752.03	113	83	100	94	108
ds-07 (800 MB, 2493 files)	800	119	75	86	86	144
combo-02 (848 MB, 382 files)	848	115	85	90	96	126
ds-08 (1126 MB, 6680 files)	1126.40	167	105	95	86	172
ds-12 (1126 MB, 2495 files)	1126.40	154	97	97	81	167
combo-07 (1137 MB,2293 files)	1137	183	111	112	112	207
ds-09 (1174 MB,18680 files)	1173.86	208	152	107	105	188
combo-04 (1178 MB,19680 files)	1178	171	115	112	107	194
combo-08 (1435 MB,21262 files)	1435	261	143	143	125	270
ds-10 (1591 MB, 1917 files)	1591.03	238	138	151	133	265
combo-09 (1648 MB,2875 files)	1648	231	144	139	133	281
combo-03 (1926 MB,4988 files)	1926	267	160	141	128	356
combo-05 (2765 MB,20597 files)	2765	458	314	218	211	680
combo-11 (3142 MB,4676 files)	3142	428	243	239	227	—
combo-10 (3426 MB,27855 files)	3426	479	295	237	212	840
combo-12 (3843 MB,11092 files)	3843	498	345	265	245	805

5.2 Best Parameter per Method

Table 6: Parameter value achieving the best elapsed time, selected representative datasets.

Dataset	Hadoop Base	Hadoop InMapper	PySpark	Java Spark
ds-00 (0.43 MB)	$r = 4$	$r = 2$	$p = 8$	$p = 4$
ds-11 (96 MB)	$r = 2$	$r = 4$	$p = 8$	$p = 24$
ds-02 (261 MB)	$r = 2$	$r = 2$	$p = 24$	$p = 24$
ds-05 (500 MB)	$r = 2$	$r = 4$	$p = 16$	$p = 40$
ds-08 (1126 MB)	$r = 4$	$r = 8$	$p = 16$	$p = 32$
ds-09 (1174 MB)	$r = 8$	$r = 8$	$p = 24$	$p = 32$
ds-10 (1591 MB)	$r = 16$	$r = 2$	$p = 16$	$p = 32$
combo-03 (1926 MB)	$r = 8$	$r = 2$	$p = 40$	$p = 40$
combo-05 (2765 MB)	$r = 2$	$r = 16$	$p = 32$	$p = 40$
combo-10 (3426 MB)	$r = 2$	$r = 1$	$p = 40$	$p = 24$
combo-12 (3843 MB)	$r = 2$	$r = 2$	$p = 32$	$p = 32$

Hadoop InMapper achieves its best time at low reducer counts ($r = 1$–4 in most cases) because the bulk of its work is done in the map phase; additional reducers add scheduling overhead without reducing shuffle cost. Both Spark engines tend to improve with higher partition counts as dataset size grows, typically settling between $p = 24$ and $p = 40$ on datasets above 1 GB, with $p = 32$ being the most frequent optimum. Hadoop Base shows no single consistently best value, indicating its bottleneck is shuffle volume rather than reducer parallelism.

5.3 Memory Usage

Memory is reported using two complementary metrics:

- `system_used_max_gb`: peak combined OS memory across all three nodes during a job (summed from 5-second samples). Reflects real cluster-wide memory pressure.
- **Driver RSS (MB)**: peak resident set size of the client/driver process on `namenode` (from `/usr/bin/time -v`). Reflects client-side overhead only.

The YARN allocation figure (`max_yarn_allocated_gb`) is not used as a primary metric because it is capped at approximately 4–5 GB by the fixed executor configuration regardless of dataset size, and therefore does not reflect workload variation.

Table 7: Peak combined system memory across all three nodes (GB), mean over all parameter values, per dataset and method. -- = OOM.

Dataset	H.Base	H.InMap	PySpark	J.Spark	Seq.Py
ds-00 (0.43 MB)	2.76	2.88	3.03	3.03	0.00
ds-01 (4.14 MB)	2.77	2.74	3.17	3.06	0.00
combo-06 (4.57 MB)	2.89	2.64	3.20	3.09	0.00
ds-11 (96 MB)	3.14	3.24	3.24	3.20	2.45
ds-02 (261 MB)	3.82	3.82	3.32	3.66	2.57
ds-03 (297.89 MB)	3.87	3.88	3.68	3.84	2.74
ds-04 (339 MB)	3.88	3.95	3.49	3.70	2.81
ds-05 (500 MB)	3.21	3.49	3.56	4.08	3.19
combo-01 (559 MB)	3.86	3.95	3.55	4.29	3.11
ds-06 (752 MB)	4.08	4.08	3.80	4.09	3.64
ds-07 (800 MB)	4.07	4.05	3.82	4.49	3.87
combo-02 (848 MB)	4.11	3.98	3.96	4.26	3.78
ds-08 (1126 MB)	3.92	3.77	3.67	4.30	2.94
ds-12 (1126 MB)	3.79	3.78	3.69	4.33	3.25
combo-07 (1137 MB)	3.87	3.92	3.92	4.56	4.12
ds-09 (1174 MB)	4.01	4.33	4.06	4.73	3.04
combo-04 (1178 MB)	3.99	3.92	3.97	4.73	3.04
combo-08 (1435 MB)	3.94	3.93	4.36	4.85	3.54
ds-10 (1591 MB)	4.63	4.88	4.32	4.77	5.03
combo-09 (1648 MB)	4.11	4.24	4.58	4.70	5.35
combo-03 (1926 MB)	4.00	3.97	4.05	4.64	4.90
combo-05 (2765 MB)	4.79	4.87	4.91	5.63	5.89
combo-11 (3142 MB)	4.21	4.31	5.02	5.54	—
combo-10 (3426 MB)	4.22	4.19	4.70	5.43	4.09
combo-12 (3843 MB)	4.58	4.96	4.69	5.62	6.33

Table 8: Peak driver/client process RSS (MB), mean over all parameter values, per dataset and method. For sequential Python this is the entire computation, not just submission overhead.

Dataset	H.Base	H.InMap	PySpark	J.Spark	Seq.Py
ds-00 (0.43 MB)	321	336	536	514	13
ds-01 (4.14 MB)	344	327	557	547	30
combo-06 (4.57 MB)	326	329	572	524	30
ds-11 (96 MB)	334	333	568	516	357
ds-02 (261 MB)	331	325	592	502	578
ds-03 (297.89 MB)	331	342	577	530	714
ds-04 (339 MB)	318	316	549	516	760
ds-05 (500 MB)	320	328	560	531	1,238
combo-01 (559 MB)	343	326	597	533	1,156
ds-06 (752 MB)	332	338	559	514	1,929
ds-07 (800 MB)	321	341	570	552	1,990
combo-02 (848 MB)	320	322	568	509	2,109
ds-08 (1126 MB)	376	362	662	594	933
ds-12 (1126 MB)	313	329	578	532	1,299
combo-07 (1137 MB)	311	345	559	510	2,189
ds-09 (1174 MB)	556	526	858	768	1,056
combo-04 (1178 MB)	541	583	880	826	1,062
combo-08 (1435 MB)	575	530	914	856	1,572
ds-10 (1591 MB)	327	330	563	522	3,181
combo-09 (1648 MB)	331	322	554	531	3,736
combo-03 (1926 MB)	353	337	610	552	3,144
combo-05 (2765 MB)	543	560	880	813	4,005
combo-11 (3142 MB)	341	338	558	561	4,751
combo-10 (3426 MB)	640	634	987	967	2,111
combo-12 (3843 MB)	446	436	668	688	4,490

6 Discussion

Hadoop InMapper vs. Hadoop Base. Across all 25 datasets, Hadoop InMapper is consistently faster than Hadoop Base, typically by 40–60%. The in-mapper combining eliminates repeated intermediate key-value pairs for the same word within each file, directly reducing the volume of data that must be sorted, shuffled, and processed by reducers. This advantage holds at every dataset size tested, from 0.43 MB to 3.84 GB.

Spark vs. Hadoop InMapper. At small and medium scales (below ≈1 GB), Hadoop InMapper is the fastest distributed method, beating both Spark engines. Fixed Spark startup overhead — container negotiation and JVM initialisation — takes approximately 11–22 seconds before any data is processed, versus 5–10 seconds for Hadoop, and this gap is a significant fraction of total runtime on short jobs. Above ≈1 GB, however, both Spark engines overtake Hadoop InMapper: at 1126 MB, Java Spark (86 s on ds-08) already beats InMapper (105 s); at 3.84 GB (combo-12), Java Spark (245 s) is 100 seconds faster than InMapper (345 s). This crossover indicates that Spark's DAG-based execution and in-memory shuffling become more efficient than Hadoop's disk-backed shuffle as dataset size grows on this cluster.

PySpark vs. Java Spark. Java Spark is consistently faster than PySpark once dataset size exceeds approximately 1 GB, while the two are nearly identical on smaller datasets where startup overhead dominates for both. At 3.84 GB, Java Spark (245 s) is 20 seconds faster than PySpark (265 s); at 2.76 GB, Java Spark (211 s) vs. PySpark (218 s). The difference is attributable to the Py4J serialization bridge and Python worker process overhead present in PySpark but absent in the native JVM execution of Java Spark. This overhead also manifests in memory: Java Spark consistently records higher cluster-wide system memory (up to 5.62 GB at 3.84 GB) than PySpark (4.69 GB at the same scale), suggesting that Java Spark holds more data in JVM heap simultaneously, which reduces serialization cost but increases memory pressure.

Effect of parallelism parameter. Increasing the reducer or partition count does not uniformly help. For Hadoop InMapper, the optimum is typically at low reducer counts ($r = 1$–4), because the map phase already performs most of the aggregation; more reducers add scheduling overhead without reducing total work. For both Spark engines, higher partition counts ($p = 24$–40) consistently help on larger datasets, but the gains plateau beyond $p = 24$–32: $p = 32$ and $p = 40$ occasionally outperform $p = 24$ by a few seconds on the largest datasets, but the difference is small and not always consistent.

Memory usage shows even less sensitivity to the parallelism parameter than elapsed time: the mean system RAM for a fixed dataset varies by only 0.2–0.5 GB across the entire r or p range. This indicates that parallelism is primarily a lever for speed, not for memory footprint, on this cluster.

Sequential Python. The sequential baseline is fastest on the smallest datasets (where distributed coordination overhead dominates) and competitive up to approximately 500 MB. Above 1 GB it is consistently slower than all distributed methods, and at 3.1 GB it fails entirely due to out-of-memory — the dictionary-based index must fit entirely in the process's address space, with no spill-to-disk mechanism. Its driver RSS grows almost linearly from 13 MB (0.43 MB dataset) to 4,751 MB (3.14 GB dataset), confirming that the full computation lives in a single process. By contrast, Hadoop's driver RSS remains almost flat across the same range (321–640 MB), since the bulk of Hadoop's memory-intensive work happens in distributed containers rather than the client process.

7 Conclusion

We implemented and evaluated five solutions for inverted-index construction across 25 datasets ranging from 0.43 MB to 3.84 GB, including 12 combined datasets created without any physical data duplication on HDFS. All 625 individual jobs were run under a single, fixed cluster configuration; 624 completed successfully. The principal findings are:

- **Below ≈1 GB:** Hadoop InMapper is the fastest distributed method. In-mapper combining reduces shuffle volume more than framework choice matters at this scale.
- **Above ≈1 GB:** Both Spark engines overtake Hadoop InMapper, with Java Spark becoming the fastest method overall on the largest datasets (>2.7 GB). Spark's in-memory shuffle becomes more efficient than Hadoop's disk-backed shuffle as data volume grows.
- **PySpark vs. Java Spark:** Java Spark is faster and more memory-intensive at scale, due to native JVM execution eliminating the Py4J serialization overhead present in PySpark.
- **Parallelism:** The partition/reducer parameter primarily controls speed; memory footprint is driven by dataset size and is largely insensitive to parallelism level.
- **Sequential Python:** Useful as a baseline for small inputs; fails entirely at 3.1 GB. Its single-process memory consumption grows linearly with dataset size, unlike the flat driver RSS of the Hadoop methods.

The project shows that reducing shuffle volume (in-mapper combining) can matter more than framework choice at small to medium scale, but that this advantage does not hold unconditionally: a well-tuned Spark implementation overtakes a carefully optimised Hadoop in-mapper-combining solution as data volume grows on a real three-node cluster.

8 Runtime Comparison Plots

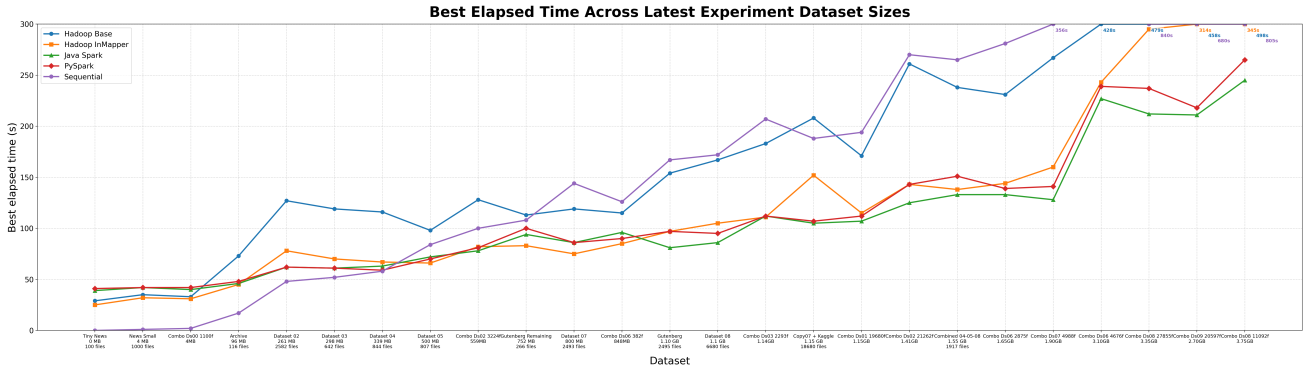


Figure 1 — Best elapsed time vs. dataset size (all 25 datasets, 5 methods).
X-axis: dataset size in MB (log scale, 0.43 to 3843). Y-axis: best elapsed time in seconds. One line per method: Hadoop Base, Hadoop InMapper, PySpark, Java Spark, Sequential Python. Expected to show: (a) sequential is fastest below ≈100 MB; (b) InMapper is best distributed method below ≈1 GB; (c) both Spark engines cross below InMapper above ≈1 GB; (d) sequential grows steeply and fails at 3.14 GB.

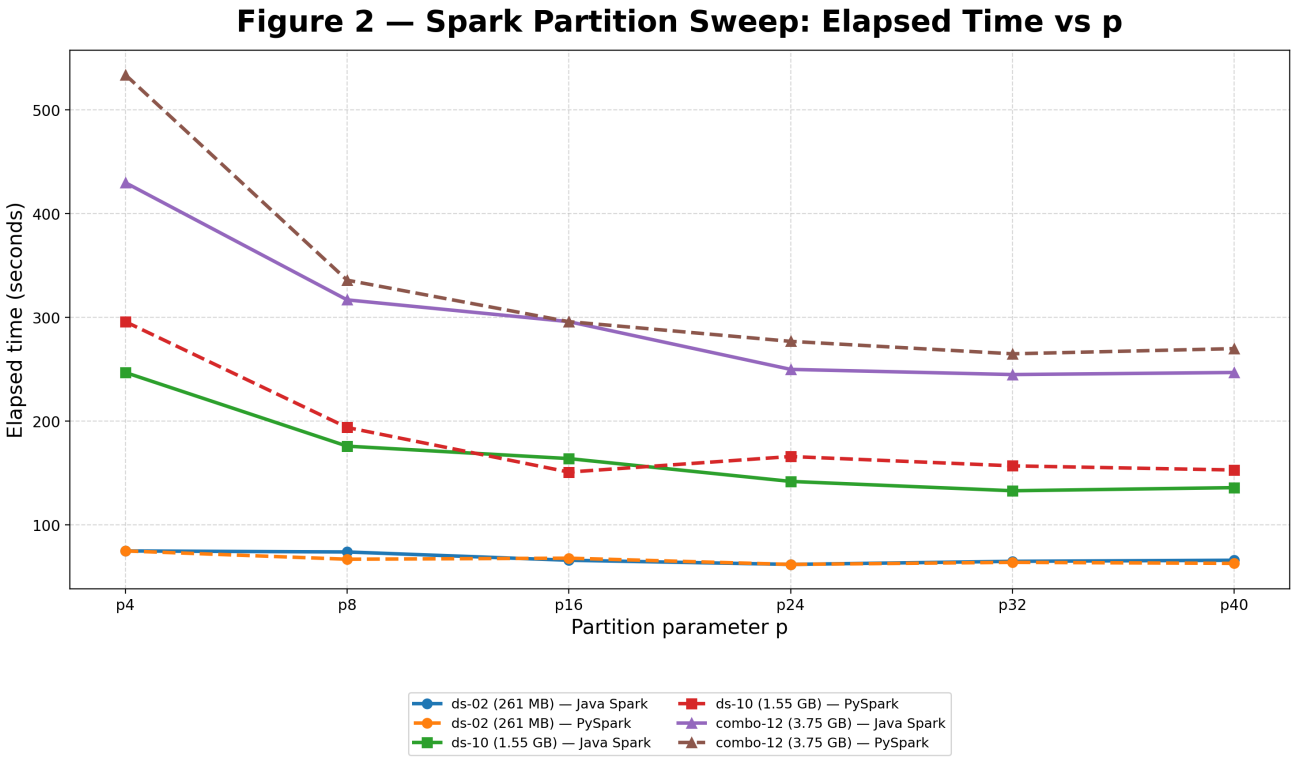


Figure 2 — Spark partition sweep: elapsed time vs. p for PySpark and Java Spark.
Three subplots or line-styles for three representative sizes: ds-02 (≈261 MB), ds-10 (≈1591 MB), combo-12 (≈3843 MB). X-axis: $p \in \{4, 8, 16, 24, 32, 40\}$. Y-axis: elapsed time in seconds. Two lines per subplot: PySpark (dashed) and Java Spark (solid). Expected to show: (a) plateau or reversal beyond $p = 24$–32; (b) Java Spark consistently below PySpark at large scale; (c) both engines nearly equal at small scale.

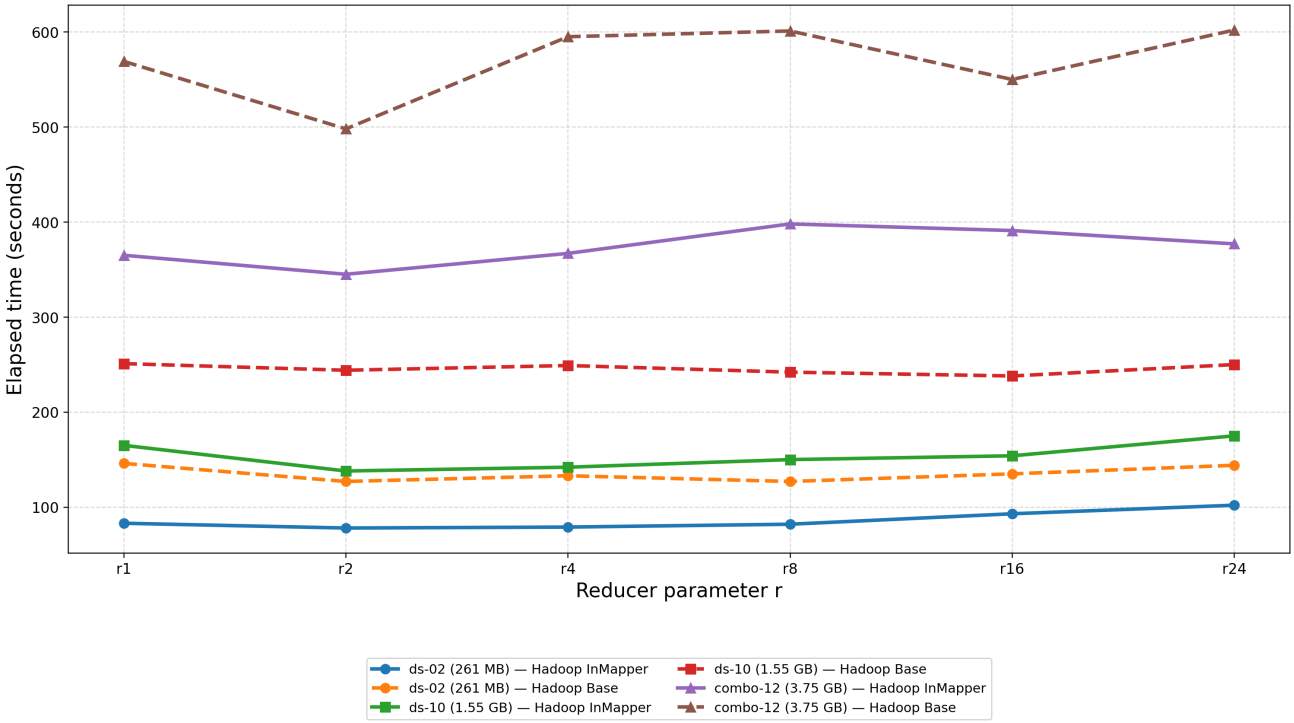


Figure 3 — Hadoop reducer sweep: elapsed time vs. r for Hadoop Base and InMapper.
Same three representative sizes as Figure 2. X-axis: $r \in \{1, 2, 4, 8, 16, 24\}$. Y-axis: elapsed time in seconds. Two lines per subplot: Hadoop Base (dashed) and Hadoop InMapper (solid). Expected to show: (a) InMapper consistently below Base at all reducer counts; (b) InMapper optimum typically at $r = 1$–4 while Base shows a flatter, sometimes non-monotonic curve; (c) the gap between Base and InMapper widens with dataset size.

9 Memory Usage Plots

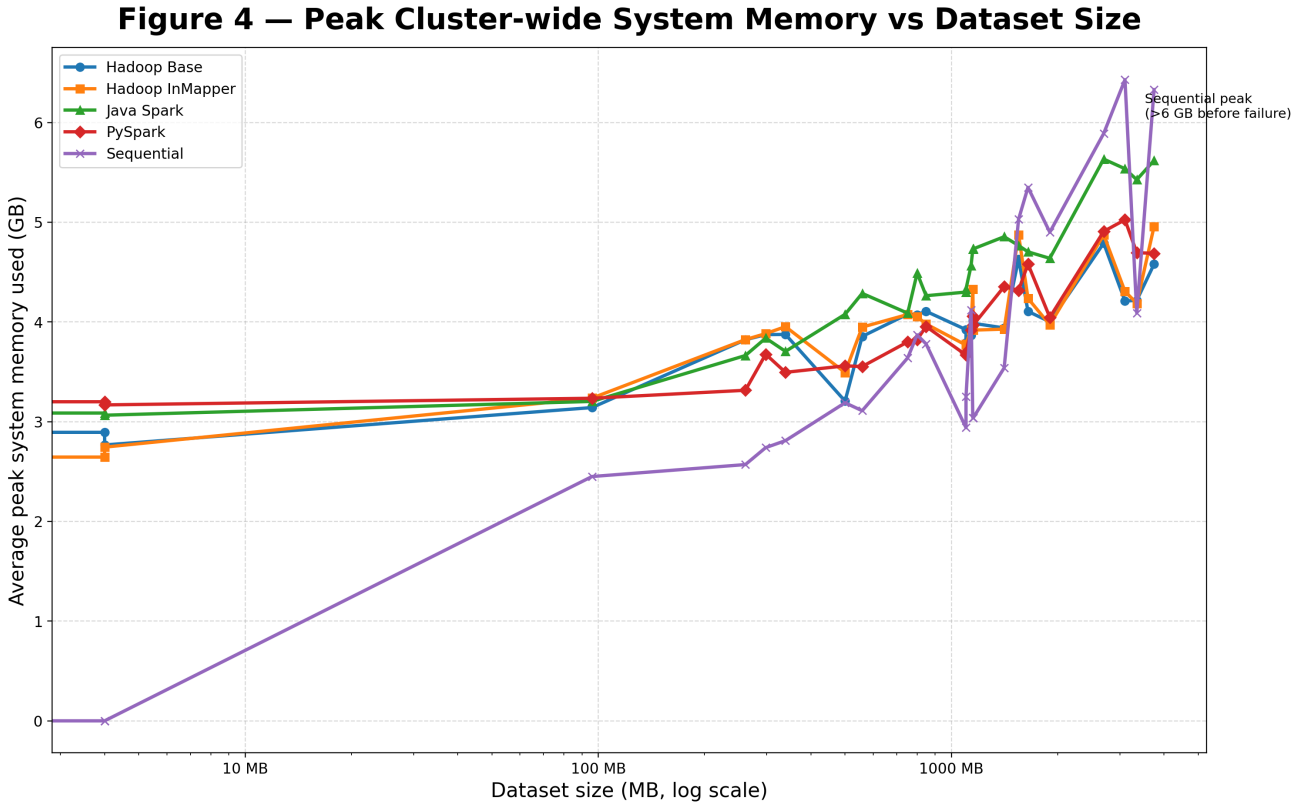


Figure 4 — Peak cluster-wide system memory vs. dataset size (all 25 datasets).
X-axis: dataset size in MB (log scale). Y-axis: `system_used_max_gb` (GB), averaged over all parameter values per dataset/method. One line per method (5 lines). Data from Table 7. Expected to show: (a) all methods scale upward with data size, confirming real memory pressure (not a flat artefact of the fixed YARN allocation); (b) Java Spark has the steepest slope (≈3.0 GB at 0.43 MB to ≈5.6 GB at 3.84 GB); (c) Hadoop methods grow more slowly and remain below Spark at large scale; (d) sequential Python is the most variable and reaches the highest values (>6 GB) before failing.

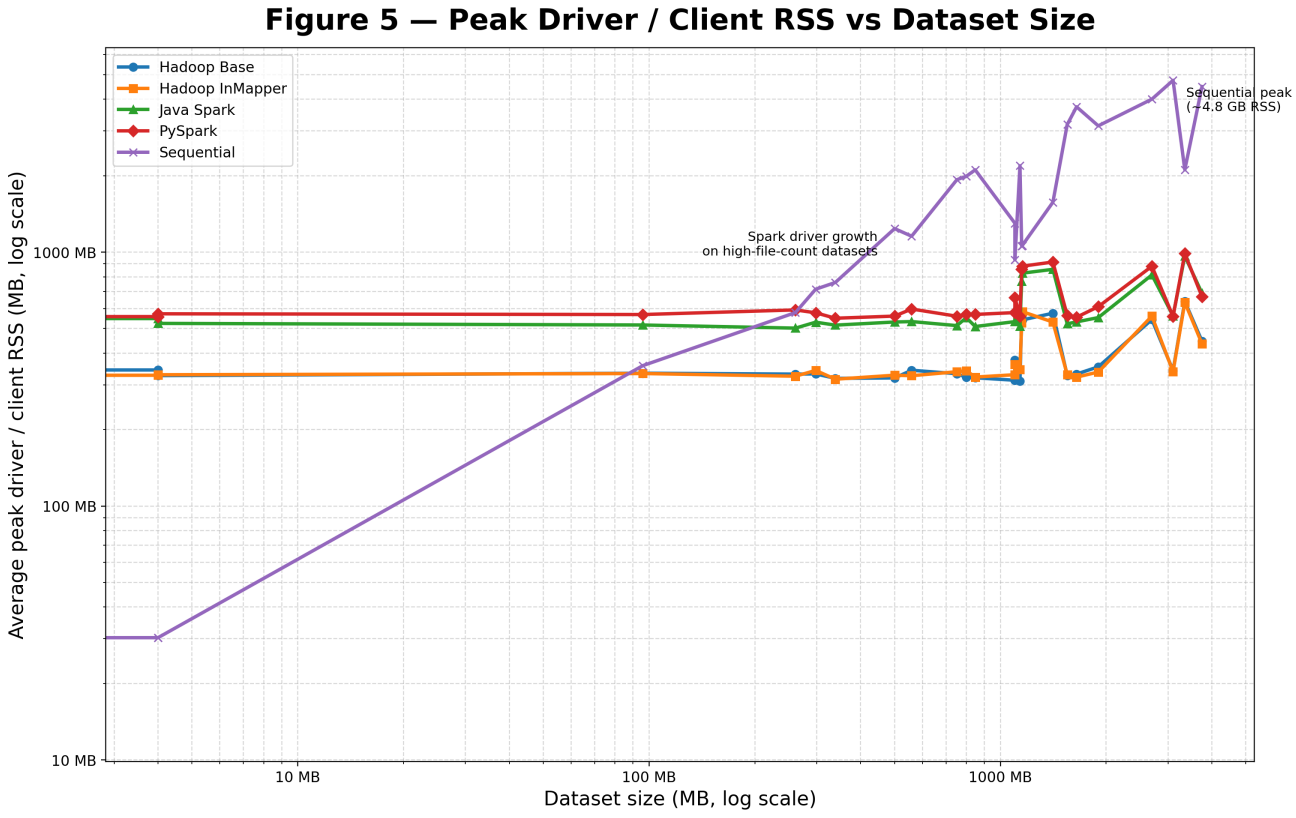
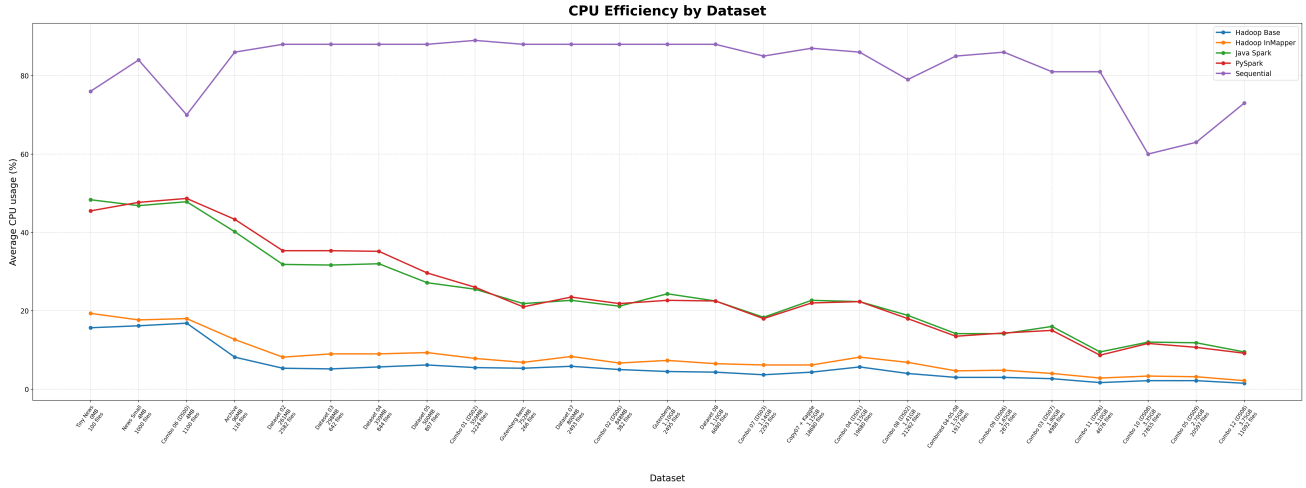


Figure 5 — Peak driver/client RSS vs. dataset size (all 25 datasets).
X-axis: dataset size in MB (log scale). Y-axis: driver RSS in MB (log scale recommended for sequential Python). One line per method. Data from Table 8. Expected to show: (a) Hadoop Base and InMapper are nearly identical and nearly flat (321–640 MB) — driver RSS does not reflect dataset size because computation happens in distributed containers; (b) PySpark and Java Spark driver RSS is higher (502–987 MB) due to JVM, Py4J, and Spark driver bookkeeping, and grows moderately with dataset size especially for high-file-count datasets (ds-09, combo-04, combo-08, combo-10); (c) sequential Python RSS grows almost linearly from 13 MB to 4,751 MB, confirming the entire computation lives in one process.



Key Findings Summary

This section summarises the principal experimental findings with respect to each requirement stated in the project specification.

1. Hadoop vs. Spark comparison (core requirement)

Both a Hadoop solution and a Spark solution were implemented and evaluated across 25 datasets (0.43 MB to 3.84 GB). The comparison shows a clear size-dependent crossover:

- **Below ≈260 MB**: Sequential Python is fastest overall; among distributed methods, Hadoop InMapper leads. Spark startup overhead (11–22 s) is the dominant cost at this scale.
- **260 MB to ≈1 GB**: Hadoop InMapper remains the fastest distributed method in most cases. Java Spark begins to match or narrowly beat InMapper on some datasets in this range (e.g. ds-02 at 261 MB: Java Spark 62 s vs. InMapper 78 s).
- **Above ≈1 GB**: Both Spark engines consistently outperform Hadoop InMapper. On the largest dataset (combo-12, 3.84 GB): Java Spark 245 s, PySpark 265 s, InMapper 345 s, Hadoop Base 498 s, Sequential 805 s.

Answer to the specification: Hadoop is more efficient than Spark at small scale due to lower fixed overhead; Spark overtakes Hadoop as data volume grows, because its in-memory shuffle becomes more efficient than Hadoop's disk-backed shuffle on this cluster.

2. In-mapper combining (extra requirement)

Hadoop InMapper outperformed Hadoop Base on every single one of the 25 datasets, with a mean speedup of **1.53**× (range 1.06×–1.83×). The advantage grows with dataset size: at 3.84 GB, InMapper (345 s) is 31% faster than Base (498 s). The benefit comes from emitting one record per unique `word@filename` pair per mapper rather than one record per token, directly reducing shuffle volume and reducer load.

Key insight: in-mapper combining is more effective than simply adding more reducers. Across all 25 datasets, InMapper's best time is always at a low reducer count ($r = 1$–4 in most cases), because the map phase already performs most of the aggregation; additional reducers add scheduling overhead without reducing the total work.

3. Sequential Python non-parallel baseline (extra requirement)

The sequential baseline was evaluated on all 25 datasets. It is the fastest method on datasets below ≈100 MB (e.g. ds-00: 0 s, ds-01: 1 s) because it has zero distributed coordination overhead. Above 1 GB it falls behind all distributed methods, and at 3.14 GB (combo-11) it was killed by the OS out-of-memory killer (exit status 137) after 423 seconds, as the in-memory Python dictionary exceeded available RAM with no spill-to-disk mechanism. Its driver RSS grows almost linearly from 13 MB to 4,751 MB across the tested size range, while the Hadoop driver RSS remains flat at 321–640 MB.

Answer to the specification: the non-parallel solution is viable only for small inputs; it becomes slower than every distributed method above 1 GB and fails entirely above ≈3 GB.

4. Effect of increasing reducer / partition count (extra requirement)

Increasing the reducer count for Hadoop and the partition count for Spark does not uniformly improve performance:

- **Hadoop InMapper**: mean elapsed time is minimised at $r = 2$ (129.6 s) and $r = 4$ (130.5 s) averaged across all 25 datasets; $r = 24$ is consistently the worst (152.5 s). Low reducer counts are optimal because in-mapper aggregation has already minimised shuffle volume.
- **Spark**: both engines benefit from higher partition counts on large datasets, typically settling between $p = 24$ and $p = 40$. However, the gains plateau beyond $p = 32$: $p = 40$ improves over $p = 24$ by only a few seconds on the largest datasets, and occasionally regresses slightly. Memory usage is almost insensitive to the partition count (variation $<$0.5 GB across the full p range for a fixed dataset), confirming that parallelism affects speed but not memory footprint.

5. Execution time and memory statistics (core requirement)

- **Execution time** was collected for all 625 job runs (5 methods $\times$ 6 parameters $\times$ 25 datasets), covering elapsed wall-clock time and best-parameter configurations as reported in Tables 5 and 6.
- **Memory** was measured at two levels: cluster-wide OS memory (`system_used_max_gb`, sampled every 5 s across all three nodes) and driver-side RSS (`max_process_rss_kb` from `/usr/bin/time -v`). Both scale measurably with dataset size, while the YARN allocation figure is flat due to the fixed executor configuration and is not used as a primary metric.
- **Java Spark vs. PySpark**: Java Spark is consistently faster ($\approx$8–33% on large datasets) but uses more system memory ($\approx$0.2–0.9 GB more at scale), due to native JVM execution eliminating the Py4J serialization overhead of PySpark.

Overall conclusion: the experiments confirm that *no single method is universally best.* The optimal choice depends on data scale: sequential Python for very small inputs, Hadoop InMapper for small-to-medium distributed workloads, and Spark (Java implementation) for large-scale processing above $\approx$1 GB. Reducing shuffle volume through in-mapper combining matters more than framework choice at small scale; at large scale, Spark's in-memory execution model overtakes even a carefully optimised Hadoop solution.